Arnab Majumder

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Education

University of Toronto Expected Apr 2028
BASc in Engineering Science, Major in Robotics Engineering *Toronto, ON*

Experience

Controls & Embedded Systems Engineer (Full-Time) Mar 2025 – Present
Aston Dynamics *Toronto, ON*
TowPilot and Delta trailer brake controllers (sold by Deutsche Hydrapro)

- Wrote the **100 Hz control loop** of a safety-critical trailer brake controller in MISRA C, later migrated to Rust, now in a **customer beta in 4 countries**.
- Moved the loop from async tasks to a deterministic **FreeRTOS** task with safety-gated outputs and non-blocking BLE commands: about 3 ms typical and **9 ms worst case** per 10 ms cycle.
- Designed a patent-pending **trailer sway controller** with **45% more damping** than the top competitor's product (ζ 0.80 vs 0.55) in ISO 9815-style steering-impulse tests at 70–140 km/h.
- Built the **sway detector** at **100% recall and 98% precision** over 100+ events: oscillation, energy and frequency checks must agree, and a hysteresis latch engages after 80 ms of sway.
- **Root-caused** a runaway to 100% braking as self-excitation (the trailer's braking feeding back into the IMU signal the controller reads) and designed compensation that held it stable from **150 km/h to a full stop**.
- Wrote the **vehicle interface** that reads any tow vehicle's lamp circuits through the 7/12-pin plug, telling braking from hazards and turn signals, including PWM-regulated truck lights.
- Wrote the state estimation in embedded Rust: an **Extended Kalman Filter (EKF)** with TRIAD calibration for IMUs mounted at any angle, and a position EKF on a from-scratch GPS parser for fleet tracking.
- Built a stacked **XGBoost + PyTorch** mass estimator (**2.3% MAPE**) that sets the brake gain automatically from signal-processed vehicle sensor data.
- **Brought up the BMI270 IMU** with a custom SPI driver in embedded Rust, debugging the sensor's startup failures on the bench.

Simulation and testing

- Built a dual-track multibody **car-trailer simulation** in C++ and Rust, validated against 8,000 km of IMU data ($\mathbf{R^2 = 0.93}$), that runs the shipped sway controller software-in-the-loop.
- Prototyped a nonlinear **Model Predictive Controller (MPC)** for sway braking that settles **2.2× faster** than the shipped controller in simulated double lane changes.
- Wrote **170+ unit tests** (requirements-based, boundary-value, fault-injection) for the control loop and reran most hardware-testable ones on the team's **hardware-in-the-loop (HIL)** rig.
- Planned and ran full-scale **vehicle testing** in Australia, tuning and verifying the controller across **8 configurations on 2 vehicles** and documenting the results.
- Co-built a wireless 100 Hz **test telemetry system** with a teammate, used for **∼500 hours**: a Rust ESP-NOW receiver, a Python FastAPI server logging to SQLite, and a live dashboard.

Battery-powered BLE product (in development, not yet public)

- Designed the electronics end to end, now in a **30-unit customer beta**: electrical architecture, part selection, a chip-down **ESP32-C3** KiCad schematic and PCB (DFM, design for cost), and bring-up.
- Wrote the firmware in C, later ported to Rust: a BLE GATT service, low-power IMU modes and software shutdown for **8–12 months** of battery life, and calibration stored in flash.
- Wrote **fault-tolerant BLE over-the-air updates** in Rust (an 8 KB ring buffer, acks and nacks that recover lost packets, a CRC check of the image) and pushed releases via AWS.

Systems Integration Engineer Sep 2023 – Mar 2025
UofT Aerospace Team – Unmanned Aircraft Systems *Toronto, ON*

- Engineered a 2-axis **antenna-tracking turret** that cut flyby link dropouts **8×** (4+ to 0.5 per hour) by aiming ahead of the drone on a path extrapolated from PX4 telemetry.

- Wrote the turret's ESP32 firmware in C++ on micro-ROS: inverse kinematics and quaternion-based PID pointing with servo-encoder feedback, simulated in Gazebo before flight.
- Developed the drone's **PX4–ROS integration** over MAVROS on its Jetson companion computer, including GPS/IMU localization, tested in PX4 simulation (Gazebo) and in flight.
- Built a 4G video pipeline (Jetson, Raspberry Pi) that raised ground-station video from **4 to 24 FPS** in flight tests.

ESROP Global Research Intern May 2024 – Aug 2024
University of Toronto & KMUTT – IC2 Lab *Bangkok, Thailand*

- Developed an **adaptive EKF** for quaternion-based IMU pose estimation that scales the accelerometer noise covariance with acceleration, trusting the gyroscope more in hard maneuvers.
- Benchmarked the filter against a standard EKF in MATLAB/Simulink and 3 maneuvering tests I designed: **32% less drift** over 10 minutes of dead reckoning on a mobile robot.

Team Captain, then Mentor Nov 2021 – Jun 2025
FIRST Robotics Competition (FRC) – FIRST Robotics Canada *Toronto, ON*

- Captained the team to its **first competition win in 8 years** and a 25-place provincial gain, then mentored new members in CAD and electrical design.
- Designed and verified the robot's **control circuits** and electrical system, chose motors and gear reductions, and tuned position, velocity and acceleration-based PID on its motor controllers.

Projects

Thrust-Vector-Controlled Rocket Sep 2024 – Present
3-Person Engineering Team *Toronto, ON*

- Designed the rocket body and **thrust-vectoring gimbal** in Onshape, and built a 3-DOF Simulink model of the rocket (Aerospace Blockset).
- Developing the GNC stack in C++: an **Unscented Kalman Filter (UKF)** feeding a PID attitude controller that kept oscillations decaying in **bench hot-fire tests**.

Deep-Learning Visual Odometry Jan 2025 – Apr 2025
Applied Fundamentals of Deep Learning, 4-Person Team *Toronto, ON*

- Built the training pipeline from EuRoC MAV ROS bags in **PyTorch**: 20 Hz camera frames synced to 100 Hz motion-capture poses, cut into **12,000** training sequences.
- Compared a vision transformer (TSFormer-VO) with a CNN-LSTM baseline (DeepVO): both drifted, but the CNN-LSTM tracked position more closely and the transformer predicted roll better.

Pet-Detection Camera Nov 2024 – Dec 2024
Personal Project *Toronto, ON*

- Built an always-on Raspberry Pi 4 camera on ROS that streams over GStreamer and Tailscale to a remote GPU.
- Ran **YOLO object detection** on the stream to spot my cat **97%** of the time and send an alert when it's in frame.

Independent Content Creator Apr 2025 – Present
Instagram – @arnabsalive *Toronto, ON*

- Grew a travel and student-life page to about **1M organic views per month** while working full-time.
- Hear from **5+ students a week** about what the content did for them.

Technical Skills

Languages: Rust, C, C++, Python, MATLAB
Controls & Estimation: Model Predictive Control (MPC), PID, EKF/UKF, sensor fusion, TRIAD calibration, vehicle dynamics, multibody simulation, linear systems analysis
Embedded: Embedded Rust, Embedded C, MISRA C, FreeRTOS, ESP-IDF, ESP32 (C3, C6), SPI, UART, I2C, CAN, BLE (GATT), ESP-NOW, OTA updates
Robotics & Simulation: ROS, ROS 2, micro-ROS, PX4, MAVROS, Gazebo, Simulink, SIL and HIL testing, Jetson, Raspberry Pi
Hardware & Tools: KiCad schematic and PCB, DFM, board bring-up, oscilloscope, multimeter, Onshape CAD, Git, Linux, CI/CD
Data & ML: NumPy, Pandas, SciPy, XGBoost, PyTorch, SQLite, FastAPI
Relevant Coursework: Control Systems, Introduction to Robotics, Mathematics for Robotics, Dynamics, Embedded Systems, Electronics for Robotics